
Sequential experimental design for overdispersed counts with Sichel distributed predictions

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Abstract

Overdispersed count rates refers to phenomena in which the count variance exceeds the average count rate. Examples include looking for diamonds in carbon-rich sedimentary layers, counting photons in gamma-ray observation, or the spike count rate in populations of neurons. We propose a procedure for sequential Bayesian experimental design under a Poisson likelihood whose rate is modulated multiplicatively with an exposure function. The prior distribution over the rate parameter is a generalized inverse Gaussian prior, which is conjugate and yields an exact posterior predictive distributed as the Sichel law. Expected information gain is then calculated from two deterministic quadrature operations, providing a faster alternative to nested Monte Carlo sampling. We show that the posterior concentrates if the accumulated exposure diverges, which is a persistence-of-excitation condition on exposure rather than on sample size. The proposed procedure is compared to state-of-the-art methods in three natural settings.

1 INTRODUCTION

Counting is fundamental to many fields of science and engineering. For example, in astronomy, photons are recorded over a period of time, and, in ecology, a camera trap counts the number of wolves passing by per week. In all these experiments, an experimenter has to choose how long the detector keeps counting, referred to as dwell time. Using the information obtained so far to optimize dwell time is called sequential opti-

mal experimental design (Lindley, 1956; Chaloner and Verdinelli, 1995). In phenomena with *overdispersed* counts, the variance of the counts exceeds the average count. For example, in gamma ray observation, the detector will often pick up barely anything but occasionally it will detect millions of photons, which means its count variance is orders of magnitude larger than the average count. The overdispersed nature of these phenomena is a challenge for sequential experimental design.

The expected information gain is an expectation of a Kullback-Leibler divergence over a predictive distribution (Ryan et al., 2016; Huan et al., 2024). In a conjugate family such as Gamma-Poisson, the predictive is negative binomially distributed which lacks the expressiveness to capture tail probabilities accurately. Outside a conjugate family, the expected information gain must be estimated by nested Monte Carlo, which is computationally expensive and ill-suited for sequential design with short dwell times (Rainforth et al., 2018, 2024). We opt to stay in a conjugate family, and will use the generalised inverse Gaussian distribution as a prior for the count rate. Its predictive is the Sichel distribution which has an additional parameter to describe tail probabilities and allows for deterministic quadrature procedures for calculating expected information gain (Sichel, 1975; Stein et al., 1987).

Related work. In active learning the same quantity is BALD (Houlsby et al., 2011), with maximum entropy sampling (Sebastiani and Wynn, 2000) dropping the aleatoric term and EPIG (Bickford Smith et al., 2023) the prediction-oriented alternative; it also appears as sensor management (Kreucher et al., 2005; Hero III et al., 2008; Yu et al., 2009), source search (Vergassola et al., 2007; Hutchinson et al., 2018) and astronomical scheduling (Loredo, 2004). Across that work expected information gain is estimated rather than computed, and those estimators carry a bias no sample size removes (Rainforth et al., 2024). We give a count family in which the quantity they approximate is available directly, taking a mixing law from the classi-

cal conjugate analysis of exponential families (Diaconis and Ylvisaker, 1979; Gutiérrez-Peña and Smith, 1997; Chen and Ibrahim, 2003) with one parameter more.

2 PROBLEM STATEMENT

Let $y \in \mathbb{N}$ be a set of counts and $\tau \in [0, \infty)$ be a dwell time. There are K contexts, with context k emitting events at an unknown rate λ_k . A design $u = (k, \tau)$ is a context and dwell time selection. Designs are chosen one at a time, until the time budget is spent. Let $\mathcal{D}_t = \{(y_i, u_i)\}_{i=1}^t$ be the designs selected and counts observed up to time t , and $\mathcal{V} = \{(y_j, u_j)\}_{j=1}^n$ be a held out test set of designs and counts.

There are three metrics of interest. First, data must be acquired such that the average surprisal on the test set is minimal. The surprisal is calculated as the negative log probability of test data under a posterior predictive distribution:

$$S = \sum_{j=1}^n -\ln p(y_j | u_j, \mathcal{D}_t). \quad (1)$$

Secondly, the model must identify as quickly as possible the context for which the count rate is maximal, k^* . That estimate is extracted from a posterior distribution over λ_k that evolves over time, i.e., $\hat{k}_t = \arg \max_k p(\lambda_k | \mathcal{D}_t)$. The goal is to minimize the regret up to time T ,

$$R = \sum_{t=1}^T |k^* - \hat{k}_t|. \quad (2)$$

Lastly, we time models on the computational cost of inference, in terms of average number of seconds to select a new design.

3 MODEL SPECIFICATION

Counts are modelled as Poisson distributed with rate parameter λ ,

$$p(y | u, \lambda_k) = \mathcal{P}(y | f(u)\lambda_k), \quad (3)$$

and exposure function $f(u)$.

Assumption 1 (Multiplicative exposure). The action acts on the rate through a known scalar $f(u)$, which does not depend on the rates.

Assumption 1 carries the analytic results below: an action decides how much is measured and not which rate, and no action renders one rate observable through another. The rates are independent under the prior and the likelihood factorises across contexts, so the rest of

this section treats a single context and writes λ for its rate.

The Fano factor is

$$F(u) = 1 + f(u) \frac{\mathbb{V}_{p(\lambda)}[\lambda]}{\mathbb{E}_{p(\lambda)}[\lambda]}. \quad (4)$$

Two properties of (4) matter. First, the departure from equidispersion is a function of the action: a Poisson process has $F = 1$ everywhere, whereas here the second term grows linearly in the exposure, so how overdispersed the observations are is itself something the design decides. Second, a negative binomial already has a free mean and variance, so overdispersion alone is no reason to leave the conjugate pair; what (5) adds is a third parameter which, with the predictive mean and variance held fixed, still moves mass between the centre and the tail, which is why the count-modelling literature reaches for the Sichel on data at once overdispersed, skewed and inflated at zero.

Write $\mathcal{GIG}$ for the generalised inverse Gaussian law,

$$\begin{aligned} p(\lambda) &= \mathcal{GIG}(\lambda | \alpha, a, b) \\ &= \frac{(b/a)^{\alpha/2}}{2K_\alpha(\sqrt{ab})} \lambda^{\alpha-1} \exp\left(- (a/\lambda + b\lambda)/2\right), \end{aligned} \quad (5)$$

with order $\alpha \in \mathbb{R}$, scale parameters $a, b > 0$, and K_α the modified Bessel function of the second kind.

Its three parameters separate once (a, b) is traded for a scale $\eta = \sqrt{a/b}$ and a concentration $\omega = \sqrt{ab}$, because the mean of (5) is then

$$\mathbb{E}_{p(\lambda)}[\lambda] = \eta \frac{K_{\alpha+1}(\omega)}{K_\alpha(\omega)}. \quad (6)$$

Requiring a particular mean fixes η in closed form and leaves α and ω free, so the brightness of a context and the shape of the uncertainty about it are set independently.

The gamma is a boundary of this family rather than a competitor: as $\omega \rightarrow 0$ with $\alpha > 0$, (5) tends to a gamma of shape α and rate $b/2$ and the marginal of (3) to a negative binomial, written $\mathcal{NB}$. The conjugate Poisson pair is therefore contained in the model, at one edge of a parameter it is otherwise free to move.

4 INFERENCE

A sequential design needs to know what a count tells it about the rate, and what to expect of the next count before committing to an action. Conjugacy of the generalised inverse Gaussian with a Poisson likelihood makes both exact (Diaconis and Ylvisaker, 1979), though neither survives evaluation in the arrangement in which it is most compactly written, which Remark 1 takes up.

4.1 Updating parameters

Under (3) and (5), a count y observed at action u yields

$$p(\lambda | y, u) = \mathcal{GIG}(\lambda | \alpha + y, a, b + 2f(u)). \quad (7)$$

Multiplying (3) by (5) and dropping every factor free of λ leaves the kernel of (5) at the stated parameters. The count enters only through the order and the action only through the second scale parameter, additively, so an exposure delivered in one measurement updates the belief exactly as the same exposure split across several. The first scale parameter a is never revised, because a Poisson likelihood carries information only through $\lambda y e^{-f(u)\lambda}$ and neither factor has the $1/\lambda$ form that a multiplies, so whatever weight the prior places on very small rates survives counting untouched. Repeating the update for n time steps gives an update of $\nu + \sum_{i=1}^n y_i$ and $b + \sum_{i=1}^n f(u_i)$, which is free of order.

A belief is therefore two numbers per context whatever the horizon, both sums, so the posterior does not depend on the order of measurement. The second is chosen by the experimenter, which raises a question a fixed design need not answer.

The Fisher information up to time t is:

$$F_t = \mathbb{E} \left[\frac{\partial}{\partial \lambda} \ln p(y | u, \lambda_k) | \lambda \right] = \sum_i i < t \frac{f(u_i)}{\lambda}. \quad (8)$$

Proposition 1 (Posterior consistency). *Fix a context with true rate $\lambda^* > 0$ and a prior (5) with $a, b > 0$ and $\alpha \in \mathbb{R}$. Since F_t is non-decreasing it has a limit $F_\infty \in (0, \infty]$, and the two cases partition the sample space. On the event $\{F_\infty = \infty\}$,*

$$\mathbb{E}_{p(\lambda | \mathcal{D}_t)} [(\lambda - \lambda^*)^2] \longrightarrow 0 \quad (9)$$

almost surely, so the posterior concentrates at λ^ . Its variance is $O(1/F_t)$, and more precisely*

$$F_t \mathbb{V}_{p(\lambda | \mathcal{D}_t)} [\lambda] \longrightarrow \lambda^* \quad (10)$$

almost surely, which is the inverse Fisher information of a Poisson observed at total exposure F_t . On the event $\{F_\infty < \infty\}$ the posterior converges almost surely to a non-degenerate law and consistency fails.

The proof is in Appendix A. Its condition falls on accumulated exposure rather than on the number of measurements, so it restricts the design and not the sample size: exposures decaying fast enough to be summable stop learning at a finite spread however many follow. This is persistence of excitation in the sense familiar from input design, the Fisher information accumulated being F_t/λ^* by (10). The unrevised a does not obstruct concentration, entering the proof only through a bracket of width proportional to a and inversely to the accumulated count, and the constant in (10) is the best a Poisson likelihood allows.

4.2 Posterior predictive distribution

Because (7) returns the family to itself, (α, a, b) denotes the current belief, prior and posterior alike, and the history is dropped from the notation.

Proposition 2 (Posterior predictive). *Under the belief $\mathcal{GIG}(\lambda | \alpha, a, b)$, the count returned by action u has mass function*

$$p(y | u, \mathcal{D}) = \frac{(f(u)\eta_u)^y}{y!} \left(\frac{\eta_u}{\eta} \right)^\alpha \frac{K_{\alpha+y}(\omega_u)}{K_\alpha(\omega)}, \quad (11)$$

where $\eta_u = \sqrt{a/(b + 2f(u))}$ and $\omega_u = \sqrt{a(b + 2f(u))}$ are the scale and concentration of the updated belief in the coordinates of (6).

Equation (11) is the Sichel law $\mathcal{Si}$ (Sichel, 1975; Stein et al., 1987) and follows from Proposition ?? without further integration, the marginal being the ratio of the two normalising constants times the rate-free factor of the Poisson mass. Every dependence on the action is carried by $f(u)$ through (η_u, ω_u) , so a grid of candidates is scored against a belief computed once; the support is unbounded, so a functional that is not a moment, the entropy in particular, is a truncated sum over it.

Remark 1 (An exact form is not an evaluation recipe). Equation (11) must not be evaluated as written. $K_{\alpha+y}$ overflows double precision at moderate order, and by Corollary ?? the order a belief carries is $\alpha + \sum_i y_i$, which grows without bound; in Section 5.1 a direct call overflows after four measurements. Every quantity is therefore accumulated in logarithms, and the run of orders a support sum needs comes by one of two routes. The recurrence $K_{\nu+1}(z) = K_{\nu-1}(z) + (2\nu/z)K_\nu(z)$ adds two positive quantities, so nothing cancels, but its rounding compounds, reaching 3×10^{-7} after 2×10^5 orders where the uniform asymptotic expansion is still exact. Short runs take the recurrence and long runs the expansion, and which is needed is set by the weight of the predictive tail.

4.3 Information gain per action

A candidate design is scored by the mutual information between λ and y under the current belief (Lindley, 1956; Houlsby et al., 2011),

$$G(u) = \mathbb{E}_{p(y | u, \mathcal{D})} [D_{\text{KL}}[p(\lambda | y, u, \mathcal{D}) \| p(\lambda | \mathcal{D})]] \quad (12)$$

$$= H[p(y | u, \mathcal{D})] - A(u),$$

the second line being the usual rearrangement into a predictive entropy less an expected conditional one,

$$A(u) = \mathbb{E}_{p(\lambda | \mathcal{D})} [H[\mathcal{P}(y | f(u)\lambda)]] \quad (13)$$

Under (3) and (5) both terms are integrals against known densities and neither needs a sample: the first

is a sum of (11) over the support of the predictive, the second a one-dimensional integral of the Poisson entropy against the belief. Neither is closed form, the support being unbounded and the Poisson entropy having no elementary expression, but both are *deterministic*, so (12) carries no Monte-Carlo error, no sample size to tune and no nesting bias, which is what an estimator outside a conjugate family must accept (Rainforth et al., 2018, 2024). Both are cheap because Proposition 2 puts every dependence on the action in $f(u)$ and (13) integrates over one positive variable. The support sum is sized from (??) and (??) and widened until the tail mass outside meets a tolerance of 10^{-9} , a normal-tail estimate not being conservative here; the second term is a 64-node Gauss-Legendre rule in $\log \lambda$. Both accumulate in logarithms, for the reason Remark 1 gives.

The decomposition in (12) separates the criteria compared in Section 5: EPIG (Bickford Smith et al., 2023), local D-optimality (Chaloner and Verdinelli, 1995), Neyman allocation (Neyman, 1992), uncertainty sampling in a total and an epistemic form (Griffiths et al., 2019), Thompson sampling, maximum entropy sampling (Sebastiani and Wynn, 2000), and the belief-free systematic and random designs. Maximum entropy sampling is (12) with $A(u)$ dropped, and a Poisson entropy grows with its mean, roughly as $\frac{1}{2} \log(2\pi e f(u)\lambda)$, so dropping $A(u)$ rewards an action for the noise it collects instead of the uncertainty it removes.

Remark 2 (What a two-moment surrogate can and cannot see). Local D-optimality scores an action by $\log(1 + f(u)\mathbb{V}[\lambda]/\mathbb{E}[\lambda])$, the log of the Fano factor (4), so it reads the belief through one scalar. Carrying that approximation into both terms of (12) gives

$$G(u) = \frac{1}{2} \log(1 + f(u)\mathbb{V}[\lambda]/\mathbb{E}[\lambda]) + c + o(1) \quad (14)$$

as $f(u)\mathbb{E}[\lambda] \rightarrow \infty$, with $c = \frac{1}{2}(\log \mathbb{E}[\lambda] - \mathbb{E}[\log \lambda]) - \Delta$ and Δ the entropy deficit of the predictive against a Gaussian of its variance. The first term is the D-optimality score exactly and c depends on the belief but not the action, so the two are an affine transform of one another along the exposure axis: contexts whose priors share a shape share c and are ranked identically by both, whatever their means and variances. Numerically Δ varies little across shapes, so c moves almost entirely through the log-mean-minus-mean-log gap, a dispersion on the log scale the Fano factor cannot see.

4.4 The Policy

The criterion (12) ranks actions but does not say what to do, because actions differ in what they cost: a budget spent in gravel or telescope time is not one counted in measurements, and criteria compared at equal numbers of measurements would favour whichever asks for

the largest sample. We therefore score an action by the information expected per unit of budget consumed, and act greedily,

$$u_{t+1} \in \arg \max_{u \in \mathcal{U}} \frac{G(u)}{c(u)}, \quad (15)$$

where $c(u) > 0$ is what the action costs. We evaluate (15) over the grid of contexts and exposures, take the winner, update that context's belief by Proposition ?? and repeat; the same normalisation is applied to every criterion in Section 5, ours included. Because the belief factorises an observation leaves the other $K - 1$ contexts untouched, so a step costs one update and $|\mathcal{V}|$ scores against the $K|\mathcal{V}|$ a naive implementation would pay. The rule is greedy over one step, which nothing above requires; we take it because every criterion in Section 5 is greedy in the same way.

5 EXPERIMENTS

5.1 Verification on GIG-Poisson data

With rates drawn from (5) and counts from (3), correct specification is arranged and any discrepancy is the implementation's. Against 4×10^5 draws, (11) carries a support mass of one to 4×10^{-13} , the update (7) agrees with numerical Bayes to a total variation of 8×10^{-13} , and the log-ratio to the negative binomial falls as ω^2 . Remark 1 is not hypothetical: a direct call to K_ν overflows at the fourth measurement, the order having reached 465.5, while in log space the Bessel quantities stay accurate to 1.3×10^{-13} against 40-digit references.

Over four regimes, prediction error falls as n^{-1} and rate error as $n^{-1/2}$, and rescaling the posterior variance by $F_n/\mathbb{E}[\lambda^*]$ collapses all four onto the constant (10) predicts to within 2 per cent, despite prior variance-to-mean ratios spanning a factor of 50; a misspecified gamma-Poisson arm converges at the same rate, paying a constant at the start and within two per cent by the thirty-second observation (Figure ??, Table 5).

5.2 Allocating a Bulk-Sampling Programme

This is the setting the Sichel law was introduced for: it was the failure of the standard discrete laws on stone counts in clustered diamond deposits that motivated the generalised inverse Gaussian (Sichel, 1975; Stein et al., 1987), to value a deposit once samples are in hand rather than to decide which to take. A property has K blocks of unknown stone density λ_k , and an action (k, v) names a block and a volume of gravel from $\{1, 2.5, 5, 10, 20, 40\}$ cubic metres, with exposure $f(u) = vr_k$ through the plant's known recovery factor,

Table 1: Bulk sampling at 12 blocks and 240 cubic metres, mean over 48 properties. Superscripts are the paired t against our agent: negative favours ours, $|t| < 2$ is a tie, and a column is bolded only where its leader beats every other at $|t| \geq 2$, which none does here. The upper block varies the acquisition with the model fixed, the lower the mixing law with the acquisition fixed. One point of Figure 1, and an abundant one.

agent	NLPD	RMSE	regret	ms
<i>acquisition, under the GIG-Poisson model</i>				
EIG (ours)	1.486	0.178	0.016	8.9
EPiG	1.490 ^{-1.4}	0.179 ^{-0.2}	0.016 ^{-0.1}	75
D-opt.	1.487 ^{-0.4}	0.187 ^{-2.0}	0.018 ^{-0.4}	0.52
Neyman	1.502 ^{-2.9}	0.172 ^{+0.7}	0.016 ^{+0.0}	1.0
systematic	1.500 ^{-3.3}	0.221 ^{-5.4}	0.015 ^{+0.3}	< 0.001
epistemic	1.564 ^{-7.0}	0.229 ^{-3.1}	0.032 ^{-2.1}	1.6
total var.	1.576 ^{-6.2}	0.261 ^{-2.4}	0.038 ^{-2.4}	1.7
random	1.597 ^{-8.3}	0.482 ^{-4.4}	0.046 ^{-3.2}	0.006
Thompson	1.701 ^{-10.5}	0.539 ^{-6.2}	0.088 ^{-4.3}	3.6
max-ent.	1.788 ^{-12.6}	0.698 ^{-5.5}	0.118 ^{-5.3}	7.9
<i>mixing law, under the EIG acquisition</i>				
gamma-Poisson	1.499 ^{-3.3}	0.199 ^{-3.4}	0.016 ^{-0.0}	0.80
lognormal-Pois.	1.492 ^{-2.0}	0.182 ^{-1.7}	0.017 ^{-0.2}	21

which is Assumption 1 exactly. The budget is gravel rather than samples, so every criterion is scored per cubic metre. Each block’s stone field is realised once as a Poisson process in processed volume, so two policies that process the same gravel recover the same stones. We report a held-out predictive score, the grade error, and the grade forgone by committing to the four best-ranked blocks.

Table 1 reports both comparisons at twelve blocks and 240 cubic metres, and against every criterion that ignores the aleatoric term the result is decisive. Maximum entropy sampling is the worst agent in the study, behind random, touching 2.8 of 12 blocks and putting 239 of 240 cubic metres into the richest block of a representative property: scoring a sample by how unpredictable its outcome is sends the budget where it teaches least per unit of gravel, which is $A(u)$ going unsubtracted. Thompson sampling and both variance criteria fail the same way.

Against the criteria that use the same posterior as carefully the accuracy is a tie and what separates them is cost. EPiG (Bickford Smith et al., 2023) and D-optimality are indistinguishable from ours, and none separates on the coarse ranking: with one parameter per context, information about the rate and about the next count rank candidates almost identically. Holding the acquisition fixed instead, the third parameter is worth 0.012 nats against the gamma boundary of its own family and 0.006 against a lognormal, both measured where our model is correct by construction,

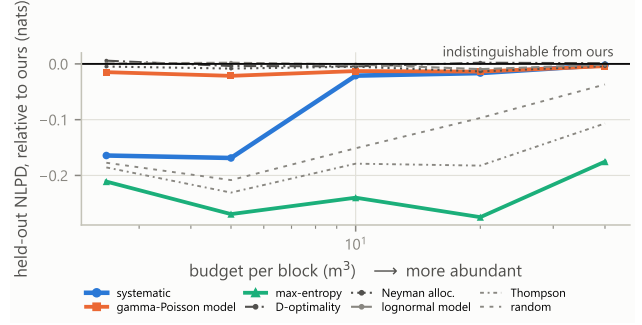


Figure 1: Predictive score relative to ours as the budget per block varies at fixed total budget, 24 properties a point; an agent on the zero line is indistinguishable from ours. The systematic programme sits on it while the budget is ample and leaves it once the budget cannot cover the blocks.

so they report what misspecification costs rather than what the parameter is worth. Appendix C shows the two effects do not interact.

Twelve blocks sharing 240 cubic metres leaves every block well sampled whatever the schedule does, so Figure 1 holds the budget fixed and varies the number of blocks it must cover. The systematic programme is indistinguishable from ours while the budget is ample, at -0.002 nats with forty cubic metres per block, and falls to -0.169 at -8 standard errors with five. The transition is sharp and sits between ten and five cubic metres per block; the choice of mixing law behaves the same way an order of magnitude smaller, the gamma falling from -0.004 nats to between -0.015 and -0.021 . An adaptive schedule is suited when the budget is scarce.

5.3 Scheduling a Gamma-Ray Observation

Every rate in the previous section is drawn from the law of Section ??, so a model assuming it is correct by construction. Here the rates are measured photon fluxes from the Fermi Large Area Telescope fourth source catalogue (Abdollahi et al., 2020), so Section 3 is the measurement process rather than an analogy: the expected count is the flux times the effective area times the time on target, so $f(u) = A_{\text{eff}} t$ and the budget is spent in megaseconds. Priors are conditioned on source class, which an observer knows before pointing and which matters, pulsars having a median flux twelve times that of blazar candidates and a variance-to-mean ratio twelve hundred times larger. The catalogue is split within each class, priors fitted to one half and sources drawn from the other.

Forty-eight targets share 120 megaseconds, so most cannot be observed at all. Under the same acquisition

Table 2: Gamma-ray scheduling at the full budget, mean over 24 programmes; conventions as in Table 1. The upper block varies the acquisition, the lower the mixing law. The two variance criteria return identical schedules here, for the reason given in the text.

agent	NLPD	dex	regret	ms
<i>acquisition, under the GIG-Poisson model</i>				
EIG (ours)	3.373	0.131	0.0066	16
EPIG	3.372 ^{+0.5}	0.130 ^{+0.6}	0.0066 ^{+0.0}	127
D-opt.	3.373 ^{+0.2}	0.130 ^{+0.5}	0.0066 ^{+0.0}	0.61
Neyman	3.379 ^{-0.6}	0.139 ^{-3.8}	0.0062 ^{+0.5}	4.0
systematic	3.783 ^{-18.7}	0.307 ^{-25.9}	0.2052 ^{-10.6}	0.001
epistemic	3.780 ^{-20.3}	0.240 ^{-21.9}	0.0621 ^{-4.2}	1.1
total var.	3.780 ^{-20.3}	0.240 ^{-21.9}	0.0621 ^{-4.2}	1.1
random	4.006 ^{-23.2}	0.344 ^{-29.0}	0.2703 ^{-8.2}	0.007
Thompson	3.943 ^{-32.1}	0.317 ^{-33.8}	0.1575 ^{-7.1}	15
max-ent.	4.143 ^{-32.5}	0.389 ^{-40.8}	0.2762 ^{-8.2}	13
<i>mixing law, under the EIG acquisition</i>				
gamma-Poisson	3.422 ^{-7.4}	0.145 ^{-4.7}	0.0098 ^{-2.3}	1.1
lognormal-Pois.	3.381 ^{-1.4}	0.131 ^{-0.0}	0.0063 ^{+0.3}	146

the gamma mixing law is 0.049 nats behind ours at -7.4 standard errors, 0.014 dex behind on the flux estimate and behind on the scheduling decision as well, the first setting here where the choice of model reaches the decision. The lognormal ties us within a standard error at nine times the cost, so what the third parameter does not buy against that alternative the conjugacy buys instead. The acquisition separates in the same regime, a systematic schedule falling 0.41 nats behind at -18.7 standard errors and maximum entropy sampling 0.77, while EPIG and D-optimality remain indistinguishable from ours.

The two variance criteria return the same schedule on every programme, which they do not in Section 5.2: the effective area is large and the fluxes heavily overdispersed, so the epistemic part of the predictive variance outweighs the aleatoric by two to four orders of magnitude and the rankings coincide, a fact about this problem rather than about the criteria.

5.4 When Does the Exact Criterion Pay?

Remark 2 says the two criteria must agree wherever contexts share a prior shape, which is how both studies above are arranged; this section varies it.

Twelve contexts fall in two classes of six, sharing a prior mean and differing in the order of the mixing law, -2 against -0.5 . The heavy class is given the *larger* index of dispersion, 1.30 against 1.00, so local D-optimality strictly prefers it while (12) strictly prefers the mild class: confident and opposite preferences, neither ever breaking a tie, which a gate verifies before the sweep runs. That direction is the interesting

Table 3: Contexts of differing prior shape, low-count regime, budget 8 or 0.67 per context, mean over 80 episodes; conventions as in Table 1. The fifth column is the share of budget given to the heavy-shape class, a mechanism rather than a score. Every agent carries the same belief, so the acquisition is all that varies. One point of Figure 2.

agent	NLPD	dex	regret	to heavy	ms
EIG (ours)	0.467	0.584	0.391	0 %	15
EPIG	0.466 ^{+0.8}	0.580 ^{+2.8}	0.383 ^{+1.3}	0 %	79
D-opt.	0.473 ^{-1.7}	0.603 ^{-5.7}	0.413 ^{-1.3}	39 %	0.72
Neyman	0.477 ^{-2.5}	0.622 ^{-9.7}	0.417 ^{-0.8}	59 %	1.2
systematic	0.476 ^{-2.5}	0.644 ^{-10.7}	0.545 ^{-4.0}	50 %	0.001
epistemic	0.482 ^{-3.3}	0.687 ^{-18.6}	0.589 ^{-5.5}	100 %	2.7
total var.	0.482 ^{-3.3}	0.687 ^{-18.6}	0.589 ^{-5.5}	100 %	2.7
random	0.477 ^{-2.4}	0.656 ^{-13.1}	0.560 ^{-4.7}	48 %	0.006
Thompson	0.483 ^{-2.7}	0.658 ^{-12.0}	0.582 ^{-5.0}	46 %	3.1
max-ent.	0.479 ^{-3.4}	0.635 ^{-11.7}	0.455 ^{-2.0}	65 %	11

one, because a large index of dispersion under a heavy-tailed mixing law is mostly unresolvable, coming from a small probability of a very large rate that no exposure recovers, and D-optimality reads that variance as something to be learned. Rates are drawn from each class’s own law and every criterion carries the same belief, so the acquisition is all that varies.

Figure 2 sweeps the budget at a low and a high count level and Table 3 reports one point of it. While the budget is scarce our criterion is ahead on the rate estimate, by 0.041 dex at -13.3 standard errors at the scarcest point and 0.019 at -5.7 with 0.67 units of exposure per context, D-optimality putting the whole budget into the heavy class and ours none of it. Lowering the counts roughly quadruples the gain at fixed budget, from -3.2 standard errors to -13.3 , which is where the approximation behind (14) is weakest.

Three things qualify this. The separation is on the rate estimate, not the held-out predictive, where D-optimality stays within two standard errors of us everywhere: (12) scores information about the rate, and the rate is what it moves. EPIG matches us throughout, so what pays is the exact predictive rather than this functional, and both need the conjugacy of Section 4. And past about 1.3 units per context the sign turns, D-optimality leading by 4.4 standard errors at 2.67: our greedy rule reinvests where the score is still high and has touched 74 per cent of contexts there against D-optimality’s hundred, so the concentration that wins under scarcity becomes under-coverage once the budget could reach everything, which is what a policy not ensuring the condition of Proposition 1 costs.

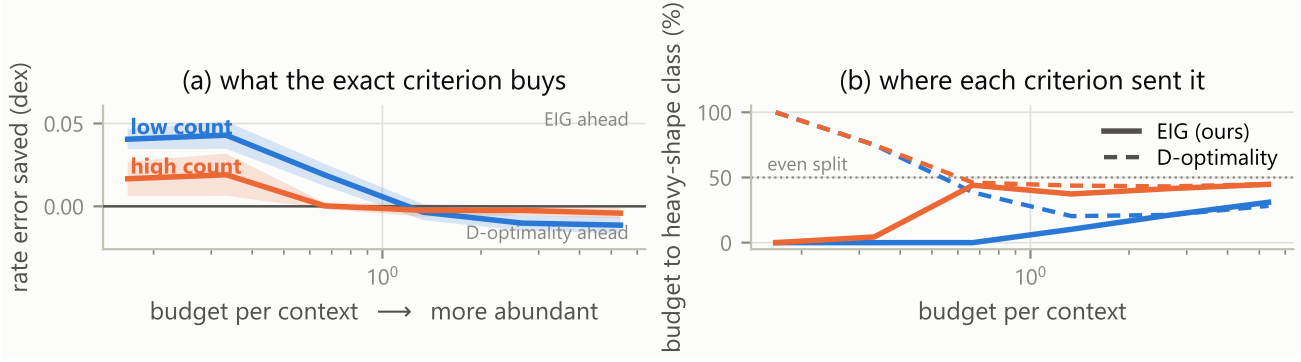


Figure 2: Twelve contexts in two classes differing in prior shape, 80 episodes a point. (a) Rate error D-optimality incurs over ours, so above zero is our gain; bands are two standard errors of the paired difference. The gain is about four times larger at low counts than at high and is gone by roughly 1.3 units of exposure per context. (b) The mechanism: D-optimality sends the whole budget to the heavy-shape class, whose dispersion is larger but mostly unresolvable, and ours none until the mild contexts are exhausted.

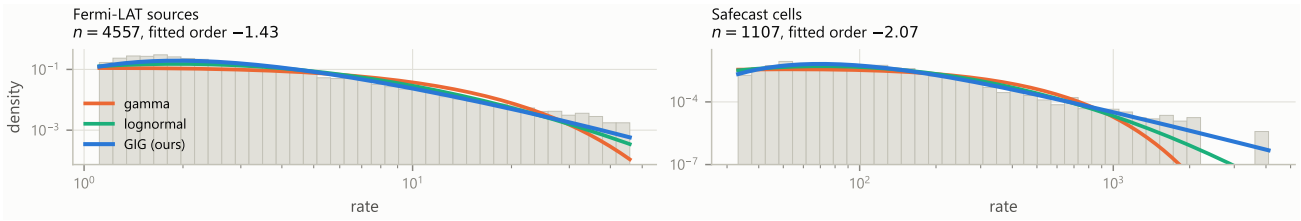


Figure 3: Three measured fields of rates against the maximum-likelihood fit of each mixing law. The gamma occupies the edge of the family at positive order, so a field fitted negative is one no gamma can reach however its two parameters are set; on the third the fitted order is positive and the two coincide.

Table 4: Three measured rate fields. ΔAIC is against the best law for that field, so zero marks the winner; the order is the generalised inverse Gaussian’s, the only law with a free one. The last column is the third parameter’s advantage over the gamma per observation, which is what compares across fields of different size.

field	n	ΔAIC			order	per obs.
		gamma	lognorm.	GIG		
Fermi-LAT	4557	1773	555	0	-1.43	0.39
Safecast	1107	568	136	0	-2.07	0.51

5.5 Which mixing law to use when?

The comparisons above either generate their rates from (5) or argue from a single field. This section asks three measured fields at once and lets them disagree: the photon fluxes of Section 5.3, the count rates of 1107 five-hundred-metre cells around Fukushima Dai-ichi recorded by the Safecast project (Brown et al., 2016), and, as a deliberate outsider, Each law is fitted by maximum likelihood with the location at zero and scored by AIC, so the third parameter pays for itself; no acquisition, budget or policy enters.

On the two physical fields the third parameter is decisive, by 1773 and 568 points of AIC against the gamma; on the linguistic field it is worth 7 points on 22460 observations, or 0.0003 each. The fitted order says why. A gamma is the boundary of the family at $\omega \rightarrow 0$ with positive order, so a field fitted at -1.43 or -2.07 is one no gamma can reach however its two parameters are set, while one fitted at $+1.73$ merely sits inside the gamma’s range, which is necessary for the gamma to suffice and not sufficient. The question can therefore be settled from the rates alone, at the cost of one fit. The ordering of the two-parameter laws is not stable: the lognormal beats the gamma on both physical fields and loses on the linguistic one.

Three features of the data qualify this. The Fermi fluxes come from a flux-limited catalogue, so the field is the detected population rather than the intrinsic one. A Safecast cell’s rate is a mean over vehicle passes spread across years of decay and neighbouring cells are not independent, so the effective sample size is below 1107. The Newsgroups field excludes terms absent from every group, removing the rare bursty terms where a heavy tail would show most, which works against the null result rather than for it.

6 DISCUSSION

Assumption 1 keeps the family closed under observation but excludes an additive term: a real gamma-ray observation counts source photons together with a known diffuse background, so the rate is $f(u)\lambda + b$, the mixture is no longer conjugate and the predictive no longer Sichel. That is the most concrete boundary of the framework we have found.

EPIG (Bickford Smith et al., 2023) and local D-optimality (Chaloner and Verdinelli, 1995) matched ours on accuracy in both allocation studies, at 0.52, 8.9 and 75 milliseconds per decision. Remark 2 says why the second ties, and it is not a fact about those problems: D-optimality is the leading term of our own criterion, so contexts whose priors share a shape are ranked identically by both, and both studies are arranged that way. Section 5.4 separates them where the theory says it should, though narrowly, and EPIG separates from us nowhere, which places the value in the exact predictive rather than in the choice of functional. Conjugacy buys an exact predictive, an acquisition free of Monte-Carlo error and a decision no two-moment surrogate reaches in one identified regime; it does not buy the cheapest rule.

Against every criterion that ignores the aleatoric term the margin is wide: maximum entropy sampling (Sebastiani and Wynn, 2000) was the worst policy in both studies, behind uniform random sampling, and the failure reproduces under every mixing law we tried, so it belongs to the criterion and not to our family.

Once the budget is large enough that every context ends up well measured, no rule distinguishes itself from visiting each in turn, and the transition is sharp, so a study reporting one budget reports one point on that curve. Past it ours is slightly behind D-optimality: Proposition 1 needs accumulated exposure to diverge at every context and Section 4.4 does not ensure it, the one place here where the theory and the policy come apart. A coverage floor would close it and we have not tested one.

The two allocation studies are simulators whose structure is taken from their domains rather than field trials; only the rate fields of Section 5.5 are measured. Contexts are independent, so a spatial problem in which neighbouring cells inform each other would break the factorisation. The rate of a context is fixed and unknown, so a process in which it is redrawn at every observation is a different model.

(Claude) That last point is unresolved and load-bearing: it decides what the right baselines are. See the memory note `overdispersion-has-two-readings`.

7 CONCLUSION

Counting experiments with a controllable exposure admit a conjugate model richer than the gamma-Poisson pair: mixing a Poisson rate over a generalised inverse Gaussian keeps the posterior in the family and makes the predictive exact, so expected information gain becomes two deterministic quadratures rather than a nested estimate, and the posterior concentrates if the design lets accumulated exposure diverge. The resulting rule beats every criterion that ignores the aleatoric term once the budget cannot cover the contexts, and is the leading-order equal of the closest two-moment surrogate except under differing prior shape, a scarce budget and low counts. Whether the third parameter earns its cost is a question about the field, answered from the rates beforehand.

AI use statement

(This section is **required** and does not count toward the page limit.)

(Wouter) Draft only. What follows is limited to AI use this repository actually records, namely drafting and revising the text and checking reported numbers against the result files. You must decide and state the rest before submission, in particular research ideation, the derivations of Section 4 and Appendix A, and the experiment code. Check the categories against the AISTATS 2027 AI Policy for Authors, since required and recommended disclosure cover different tasks.

In this work, we used generative AI tools to draft and revise the text of this manuscript, and to check the numbers reported here against the recorded outputs of the experiments. [State here whether generative AI was used for research ideation, for the derivations of Section 4 and Appendix A, and for the experiment code, together with how that work was verified.] We have reviewed all AI-assisted work. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
 - (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
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3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
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Sequential experimental design for overdispersed counts with an exact Sichel predictive: Supplementary Materials

A PROOF OF POSTERIOR CONSISTENCY

Throughout, one context is fixed and its rate is written λ , with true value $\lambda^* > 0$. Let $\mathcal{F}_t$ be the filtration generated by $(u_1, y_1, \dots, u_t, y_t)$ and let the policy be adapted, so that u_i is $\mathcal{F}_{i-1}$ -measurable and may depend on everything observed before step i . Write $f_i = f(u_i)$, and

$$T_t = \sum_{i \leq t} y_i, \quad F_t = \sum_{i \leq t} f_i, \quad \nu_t = \alpha + T_t, \quad b_t = b + 2F_t,$$

so that by Corollary ?? the posterior is $\mathcal{GI}(\lambda | \nu_t, a, b_t)$, with scale $\eta_t = \sqrt{a/b_t}$ and concentration $\omega_t = \sqrt{ab_t}$. Two identities are used repeatedly, both immediate: $\eta_t/\omega_t = 1/b_t$ and $\eta_t\omega_t = a$. The moments of the generalised inverse Gaussian are

$$\mathbb{E}[\lambda^n] = \eta_t^n \frac{K_{\nu_t+n}(\omega_t)}{K_{\nu_t}(\omega_t)}, \quad (16)$$

so everything below is a statement about the ratio $R_\nu(z) = K_{\nu+1}(z)/K_\nu(z)$.

A.1 A Bracket on the Bessel Ratio

The proof needs no asymptotic expansion of K . An exact two-sided bound follows from the recurrence alone, and it is the only property of the Bessel functions used.

Lemma 1. *For every $z > 0$ and every real ν ,*

$$R_\nu(z) > \frac{2\nu}{z}, \quad (17)$$

and if in addition $\nu > 1$,

$$R_\nu(z) < \frac{2\nu}{z} + \frac{z}{2(\nu-1)}. \quad (18)$$

Proof. The modified Bessel function of the second kind is strictly positive on $(0, \infty)$ for every real order, and satisfies $K_{\nu+1}(z) = K_{\nu-1}(z) + (2\nu/z)K_\nu(z)$. Dividing by $K_\nu(z)$ gives the exact recursion

$$R_\nu(z) = \frac{2\nu}{z} + \frac{1}{R_{\nu-1}(z)}, \quad (19)$$

valid for every real ν . Positivity of K makes $R_{\nu-1}(z) > 0$, so the second term of (19) is strictly positive and (17) follows. Applying (17) at order $\nu-1$, which is a positive lower bound exactly when $\nu > 1$, gives $R_{\nu-1}(z) > 2(\nu-1)/z$ and hence $1/R_{\nu-1}(z) < z/2(\nu-1)$. Substituting into (19) gives (18). $\square$

A.2 A Law of Large Numbers for the Accumulated Count

Lemma 2. *On the event $\{F_t \rightarrow \infty\}$, $T_t/F_t \rightarrow \lambda^*$ almost surely, and $\nu_t \rightarrow \infty$.*

Proof. Under λ^* the conditional law of y_i given $\mathcal{F}_{i-1}$ is Poisson with mean $f_i\lambda^*$, since f_i is $\mathcal{F}_{i-1}$ -measurable. Hence $M_t = T_t - \lambda^*F_t$ is a square-integrable martingale with predictable quadratic variation

$$\langle M \rangle_t = \sum_{i \leq t} \mathbb{V}[y_i | \mathcal{F}_{i-1}] = \lambda^*F_t.$$

On $\{F_t \rightarrow \infty\}$ we have $\langle M \rangle_\infty = \infty$, so the strong law for square-integrable martingales gives $M_t/\langle M \rangle_t \rightarrow 0$ almost surely, that is $T_t/F_t \rightarrow \lambda^*$. Since $\lambda^* > 0$ and $F_t \rightarrow \infty$, also $T_t \rightarrow \infty$ and therefore $\nu_t = \alpha + T_t \rightarrow \infty$. $\square$

The adaptivity of the policy is handled entirely by this lemma, and it is the only place it is needed. Nothing later distinguishes a sequential design from one fixed in advance.

A.3 Proof of Proposition 1

Proof. Work first on $\{F_\infty = \infty\}$, intersected with the almost sure event of Lemma 2, on which $\nu_t \rightarrow \infty$ and so $\nu_t > 1$ for all large t .

The posterior mean. Apply (16) at $n = 1$ and Lemma 1 at $\nu = \nu_t$, $z = \omega_t$. Using $\eta_t/\omega_t = 1/b_t$ and $\eta_t\omega_t = a$ to simplify each side,

$$\frac{2\nu_t}{b_t} < \mathbb{E}_{p(\lambda|\mathcal{D}_t)}[\lambda] < \frac{2\nu_t}{b_t} + \frac{a}{2(\nu_t - 1)}. \quad (20)$$

The width of the bracket is $a/2(\nu_t - 1) \rightarrow 0$, and the leading term satisfies

$$\frac{2\nu_t}{b_t} = \frac{2F_t}{b + 2F_t} \cdot \frac{\alpha + T_t}{F_t} \rightarrow 1 \cdot \lambda^* \quad (21)$$

by Lemma 2. Hence the posterior mean converges to λ^* almost surely. Note that the first scale parameter a appears nowhere except in the width of (20), which is the formal content of the observation that a is never revised and never needs to be.

The posterior variance. By (16), $\mathbb{V}[\lambda] = \eta_t^2 R_{\nu_t}(R_{\nu_t+1} - R_{\nu_t})$, with the argument ω_t suppressed. Applying (19) at orders $\nu_t + 1$ and ν_t and discarding the negative term $-1/R_{\nu_t-1}$,

$$R_{\nu_t+1} - R_{\nu_t} = \frac{2}{\omega_t} + \frac{1}{R_{\nu_t}} - \frac{1}{R_{\nu_t-1}} < \frac{2}{\omega_t} + \frac{\omega_t}{2\nu_t},$$

where (17) bounds $1/R_{\nu_t}$. Combining this with (18) for R_{ν_t} and expanding the product gives

$$\mathbb{V}_{p(\lambda|\mathcal{D}_t)}[\lambda] < \frac{4\nu_t}{b_t^2} + \frac{a}{b_t} \left(1 + \frac{1}{\nu_t - 1}\right) + \frac{a^2}{4\nu_t(\nu_t - 1)}. \quad (22)$$

By Lemma 2, $\nu_t/F_t \rightarrow \lambda^*$ and $b_t/F_t \rightarrow 2$, so the first term of (22) is asymptotic to λ^*/F_t , the second is of order $1/F_t$ and the third of order $1/F_t^2$. The posterior variance is therefore $O(1/F_t)$ and vanishes.

Concentration. The posterior mean-squared error decomposes as

$$\mathbb{E}_{p(\lambda|\mathcal{D}_t)}[(\lambda - \lambda^*)^2] = \mathbb{V}_{p(\lambda|\mathcal{D}_t)}[\lambda] + \left(\mathbb{E}_{p(\lambda|\mathcal{D}_t)}[\lambda] - \lambda^*\right)^2,$$

and both terms vanish almost surely by the two paragraphs above, which is (9). Concentration follows by Chebyshev's inequality, since for any $\epsilon > 0$,

$$p(|\lambda - \lambda^*| > \epsilon | \mathcal{D}_t) \leq \epsilon^{-2} \mathbb{E}_{p(\lambda|\mathcal{D}_t)}[(\lambda - \lambda^*)^2] \rightarrow 0.$$

The sharper statement (10) needs a finer bound on the same ratio and is proved in Appendix A.4.

The converse. Work now on $\{F_\infty < \infty\}$. Then $\sum_i \Pr(y_i > 0 | \mathcal{F}_{i-1}) = \sum_i (1 - e^{-f_i \lambda^*}) \leq \lambda^* F_\infty < \infty$, so by the conditional Borel–Cantelli lemma only finitely many observations are nonzero almost surely. Hence $T_t \rightarrow T_\infty < \infty$, and the posterior converges to $\mathcal{GIG}(\lambda | \alpha + T_\infty, a, b + 2F_\infty)$, which has strictly positive variance because $b + 2F_\infty < \infty$. The posterior does not concentrate, at any λ^* . $\square$

Table 5: Fitted log-log slopes over the upper half of the sample-size range, by regime. The generating parameters change the constant and not the exponent. The last column is the worst the misspecified gamma-Poisson arm does against ours over the whole range, which in every regime is at the first observation.

regime	$\mathbb{V}/\mathbb{E}$	KL	rate err.	$\mathbb{V}[\lambda \mathcal{D}_n]$	NB cost
light	0.8	-0.971	-0.492	-0.987	$\times 1.01$
moderate	4.0	-0.952	-0.472	-0.995	$\times 1.11$
heavy	40	-0.959	-0.498	-0.997	$\times 1.90$
sparse	0.5	-0.935	-0.480	-0.973	$\times 1.08$
theory		-1	-1/2	-1	

A.4 The Constant in the Rate

Bound (22) gives the order $1/F_t$ but not the sharp constant, because its second term contributes at the same order as the first. The looseness is an artefact of bracketing R_{ν_t} and $R_{\nu_t+1} - R_{\nu_t}$ separately, when the variance is a difference of two nearly equal quantities. Substituting (19) into itself once more sharpens (18) to

$$R_\nu(z) = \frac{2\nu}{z} + \frac{z}{2(\nu-1)} + O(z^3/\nu^3), \quad (23)$$

uniformly as $\nu/z \rightarrow \infty$, which holds here because ν_t/ω_t grows like $\lambda^* \sqrt{F_t/2a}$. The correction term must be kept at $\nu-1$ rather than ν , since the two differ by $O(z/\nu^2)$ and that is larger than $O(z^3/\nu^3)$ in this regime. Differencing (23) at consecutive orders cancels the correction to leading order, because $z/2\nu - z/2(\nu-1) = -z/2\nu(\nu-1)$, and leaves $R_{\nu_t+1} - R_{\nu_t} = 2/\omega_t + O(\omega_t/\nu_t^2)$. Hence

$$F_t \mathbb{V}_{p(\lambda | \mathcal{D}_t)}[\lambda] = \frac{4\nu_t F_t}{b_t^2} (1 + o(1)) \rightarrow \lambda^*, \quad (24)$$

which is the inverse Fisher information of a Poisson observed at total exposure F_t and proves (10). Simulation agrees. Across true rates $\lambda^* \in \{0.2, 3, 50\}$, orders $\alpha \in \{-3, -0.5, 2\}$ and a prior mean deliberately set to 1 in every case, the ratio $F_t \mathbb{V}[\lambda] / \lambda^*$ reaches 1.000 to 1.007 by $F_t = 10^5$.

B FIGURES AND SLOPES FROM THE STUDIES

The three figures below accompany Sections 5.1 to 5.3. Their numbers appear in Tables 5, 1 and 2 of the main text.

C THE FULL CRITERION-BY-MODEL GRID

Every criterion was run under every mixing law on the same 48 properties, so that the acquisition and the model can be varied independently. Table 6 reports the held-out predictive score of each criterion relative to expected information gain under the same model, in nats, so that each column is internally comparable and a row that is constant across columns is a criterion whose behaviour does not depend on the model carrying it.

EPIG is not run under the lognormal mixing law. Its outer expectation requires a posterior predictive entropy at every node of a quadrature over the candidate’s own predictive, and that predictive is itself a quadrature over a grid posterior when the model is not conjugate. The nesting costs more than the rest of the study combined, which is a statement about non-conjugacy rather than about the criterion, and it is recorded here rather than passed over.

D UNDERPERFORMANCE FOR MODEL MIS-SPECIFICATION

We study the relative rates of 1123 terms across the twenty corpora of 20 Newsgroups.

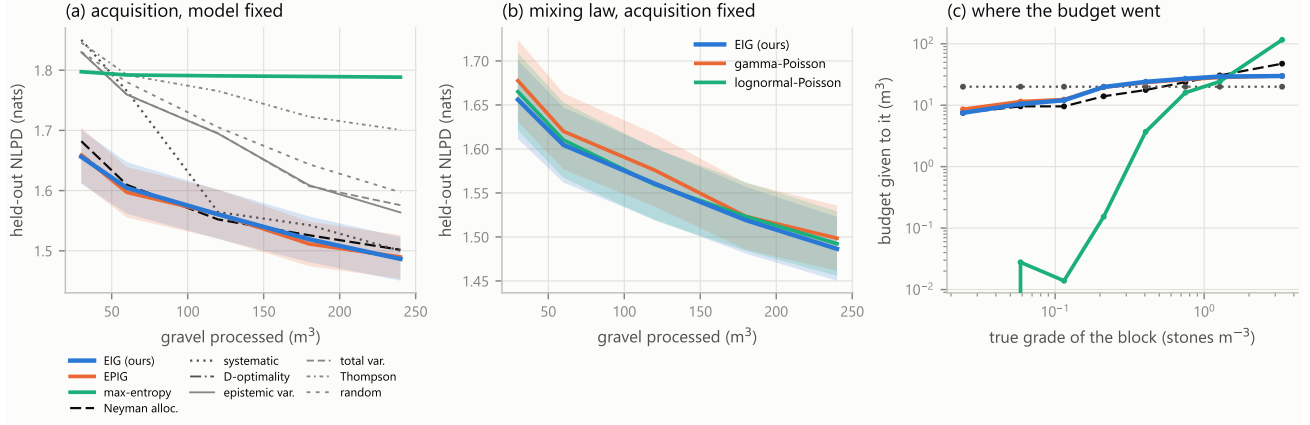


Figure 4: Bulk sampling under a processing budget, 48 properties of 12 blocks each. Bands are one standard error. (a) Holding the model fixed and varying the acquisition. (b) Holding the acquisition fixed and varying the mixing law. Note the axis, which spans a fifth of the range in (a). (c) Where the budget went, against what each block was worth. The systematic programme is flat by construction, the informed criteria tilt gently towards blocks with more to learn, and maximum entropy sampling abandons everything but the richest.

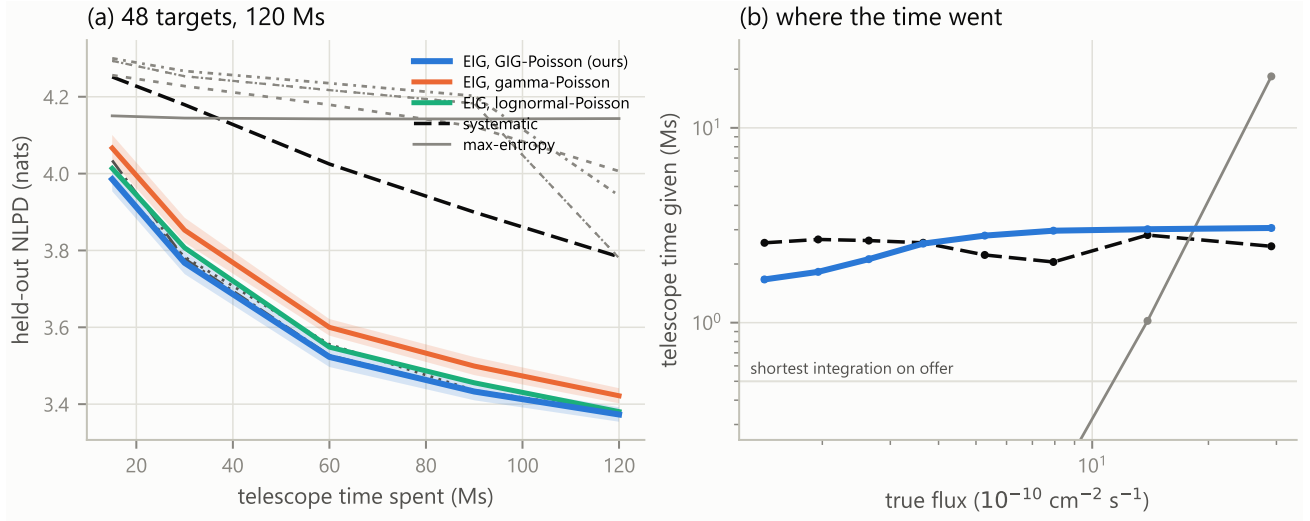


Figure 5: Scheduling telescope time across 48 gamma-ray targets on a budget of 120 megaseconds, 24 programmes. (a) Held-out predictive score against telescope time spent, with bands at one standard error. (b) Time given to a target against how bright it turned out to be. Maximum entropy sampling spends the budget on the brightest targets and the informed criteria do not. The measured flux field against each fitted mixing law is the first panel of Figure 3.

Table 6: Held-out NLPD relative to expected information gain under the same model, in nats. Positive is worse. Rows are near-constant across columns, which is the point, since the cost of the acquisition does not depend on the mixing law.

criterion	GIG-Poisson	gamma-Poisson	lognormal-Poisson
EPIG	+0.003	−0.002	not run
D-optimality	+0.001	+0.001	−0.003
Neyman allocation	+0.015	+0.018	+0.009
epistemic variance	+0.077	+0.066	+0.083
total variance	+0.089	+0.079	+0.084
Thompson sampling	+0.214	+0.219	+0.191
maximum entropy	+0.302	+0.308	+0.296